

(MSCI), S&P GSCI, Barclays Aggregate Bond Index (BOND), and US Dollar Index (USDI). The percentage changes of [Fama and French \(1993\)](#) three factors (SMB, HML), [Fama and French \(2015\)](#) five factors (RMW, CMA) as well as the risk free rate were downloaded from Kenneth French's website.⁵ Finally, the Value (VAL) and Momentum (MOM) Everywhere factors of [Asness et al. \(2013\)](#) are available from the AQR website.⁶ Data for all the above are available from January 1985 to March 2015.

2.2. Trading Signals of Trend-Following Strategies

Two types of trend-following strategies are employed in this study, TSM and RSM.⁷ Previous studies suggest that 12 months is the optimal look-back period (j) for trend-following strategies, and holding period (h) usually ranges from 1-12 months, see, among others, [Moskowitz et al. \(2012\)](#), [Zhou and Zhu \(2013\)](#) and [Papailias et al. \(2021\)](#). Hence, in this study we adopt a 12-month value of j for all the benchmark strategies.⁸

The TSM signals are generated in the same way as in [Moskowitz et al. \(2012\)](#), where a long position is indicated if the period return is positive, i.e., the annual return using $j = 12$; otherwise, the investor opens a short position on security s . The TSM returns are given as follows:

$$R_t^s | PR_{t-12,t-1}^s = \begin{cases} +r_t^s, & PR_{t-12,t-1}^s > 0 \\ -r_t^s, & PR_{t-12,t-1}^s < 0 \end{cases}, \quad (1)$$

where $PR_{t-12,t-1}^s$ is the period return of instrument s during time $t - 12$ to $t - 1$, as suggested by our look-back period $j = 12$.

The RSM signals are generated when the empirical probability of the positive

⁵http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

⁶<https://www.aqr.com/>

⁷We also examine the properties of time series reversal based on a simple moving average (SMA) trading rule, which is one of the most fundamental trend-following strategies. The results of SMA are very similar to those of TSM and RSM, and are discussed in Section B of the Online Appendix.

⁸We also have additional results based on look-back periods other than 12 months which are available upon request.